Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/28/2014
PI Name:	Chahine, Georges D.Sc.		
Project Title:	Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator		
Division Name:	Physical Sciences		
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PI Email:	glchahine@dynaflow-inc.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	301-604-3688
Organization Name:	Dynaflow, Inc.		
PI Address 1:	10621 Iron Bridge Rd, Suite J		
PI Address 2:			
PI Web Page:			
City:	Jessup	**State:**	MD
Zip Code:	20794	**Congressional District:**	7
Comments:			
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Contact Monitor:	Sharp, Lauren	**Contact Phone:**	
Contact Email:	lauren.m.sharp@nasa.gov		
Flight Program:			
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COI Name (Institution):	Wu, Xiongjun (Dynaflow, Inc.) Loraine, Greg Ph.D. (Dynaflow, Inc.)		
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	The limits on the amount of liquids and gases carried to space make it imperative to recycle and reuse. On Earth, bubbles are relatively easily separable by buoyancy. In microgravity, other external forces such as a centrifugal force must be utilized to separate bubbles from liquids. Two categories of centrifugal force separators exist: one uses active rotation by mechanically spinning the tank. This is very efficient but requires shaft, bearings, and motor. In the second, the tank is fixed and the rotation is induced by eccentric injection of the mixture (Free Vortex Separator or FVS). This FVS has been proven very efficient for mid-range void fraction (50-80% gas) with an efficiency approaching 100%. However, this FVS is not as efficient for lower void fraction, and there is a need of a separator for low void fractions. We have developed a phase separator at DYNAFLOW based on a DYNASWIRL® swirl flow nozzle. It consists of two concentric cylinders, the fluid is introduced in the space between the two cylinders, and the swirling flow inside the inner		

Task Description:	cylinder is generated with the help of tangential injection slots. With this configuration one can make the vortex circulation intense enough to generate cavitation with a very stable vaporous/gaseous core even at low flow rates. Operation with the high swirl near or slightly below cavitation inception in the vortex core guaranties removal of air even from mixtures with very low void fractions. This is achieved by forcing the bubbles to expand and collect in the vortex core. These are then removed through two two central exit orifices at each end of the cylinder. This configuration was tested successfully during a first reduced-gravity flight and we are planning additional testing this month. We are further developing and experimentally testing this configuration to characterize its gas-liquid separation performance and to improve its efficiency. We are also conducting numerical simulations to help optimize the design and ultimately build a compact system for space validation on the International Space Station (ISS) using the existing Fluid Integrated Rack facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As a passive separator device, the DYNASWIRL® phase separator has the advantage of simplicity and dependability. It requires no moving mechanical parts and low power. Compared to other passive separator devices, which usually are not very efficient at low void fraction, the DYNASWIRL® is able to force low void fraction gas out of the liquid into the central core of the vortex by combining the effects of swirl, cavitation, and rectified gas diffusion. In addition to being an effective phase separator for space applications, it will also find wide applications in Earth applications where fast separation of low void fraction gas is of concern.
Task Progress:	During this reporting period, we have continued our efforts to investigate and improve our DYNASWIRL® phase separator. We have prepared and conducted a reduced gravity test, addressed the Science Concept Review (SCR) panel's comments, and redesigned the test section to reduce its size and flow rate to fit in the new NASA breadbox test rig design that is suitable for experiments on the International Space Station (ISS). We tested the DYNASWIRL® phase separator in a NASA reduced gravity flight in September 2013 and obtained data about its operation in off-design conditions in reduced gravities. Systematic numerical simulations as well as ground based experiments were performed to evaluate and improve the new test section design. We also developed new schemes to improve liquid exit and gas extraction. Additionally, light attenuation as well as conductivity based schemes were investigated as means to measure the void fraction of the two phase mixture. A. Reduced Gravity Flight Test in September 2013. The DYNASWIRL® phase separator, which was successfully tested in reduced-gravity in May 2012, was sent to NASA to be integrated into an existing two-phase flow test rig and tested under new operating conditions in a reduced gravity flight. The purpose was to evaluate how the separator would behave in off-design conditions and in transient conditions. The separator operation conditions required a liquid flow rate of at least 10 gpm, while the test rig in which it was integrated could only supply flow rates less than 5 gpm. The reduced gravity tests were performed on September 19th and September 20th, 2013. During the reduced gravity flight tests, various combination of liquid flow rates between 2 gpm and 5 gpm and gas flow rates between 0 and 2 gpm were tested for various preset reduced gravity conditions. The flight tests provided good information about the transient behavior of the phase separator at reduced gravity and indicated that the phase separator needed to be redesigned for the lower allowable total liquid volume and lower flow rates. . B. Small-Scale Phase Separator Design. After the reduced gravity flight test in September 2013, NASA decided to build a completely new breadboard test rig as a template for the test rig to be used in International Space Station (ISS) experiments. The breadboard is being designed to be compatible with the ISS available resources of liquid, gas, and power supplies. This new test rig is significantly smaller than the test rig in which the previous versions of the DYNASWIRL® phase separator were tested and a smaller scale DYNASWIRL® phase separator is needed. For illustration, the maximum allowable total liquid volume is 1 gallon for the new rig versus 3 gallon for the old rig. Also, the maximum available liquid flow rate is 5 gpm versus 18 gpm for the old test rig. With the assistance of numerical simulations and ground experiments at DYNAFLOW, a couple of design iterations were completed in this reporting period, which resulted in a successful design within the capacity of the new test rig. The small-scale DYNASWIRL® chamber has a footprint of 6.25 in. x 7.5 in. x 7.5 in. and a total volume of 0.25 gallons. With this new design, a stable vortex core can be formed at around 1 GPM and a flow rate of about 3.5 GPM can be achieved at the nominal pump head pressure of 40 psi, all within the capabilities of the new test rig. C. Improved Liquid Exit. Due to the DYNASWIRL® chamber scale down, the exit orifices arranged in an annular fashion around the chamber axis are at a much smaller radial distance from the axis. This reduces the upper limit of the maximum vortex gaseous core size allowable for the gas to not escape into the liquid exit. In order to maintain an efficient gas extraction by allowing a larger gas core and prevent the gas from escaping the following two new design concepts were adopted: (a) a vortex blocker, which prevents the gas from escaping from the vortex core into the liquid exit even for thick gas cores, was added (b) the tangential injection slots were moved away from the liquid exit. With the vortex blocker, the vortex core can grow as large as the vortex blocker diameter without gas leaking into the liquid exits. The size of the vortex blocker and the location to place it were estimated through numerical simulation and validated through laboratory experiments. With the farther upstream tangential injection, the chances for the bubbles to be trapped in the vortex core are maximized while the risk for them to reach behind the vortex blocker and escape into the liquid exit is minimized. D. Improved Gas Extraction Scheme. To further improve gas extraction, a new gas extraction scheme was developed. The new scheme allows gas extraction along a much larger area, utilizing the full length of the gaseous vortex core instead of just from an orifice of the same size as a cross-section of the core. This results in a significant increase in the contact area for gas extraction and a much improved extraction speed especially given the reduction in the size of the separator. E. Numerical Simulations. 3D numerical simulations were conducted by coupling our 3DYNAFS-VIS©, a Navier-Stokes flow solver, which models the overall two phase flow field of the DYNASWIRL® phase separator, and 3DYNAFS-DSM©, a bubble tracking and dynamics and motion solver. Both components are modules of our 3DYNAFS© CFD suite. The simulations enabled us to better understand the gravity effects under different test section designs and flow conditions. The simulations also provide guidelines to improve the phase separator design. The results from the numerical simulation were used to study the effects of the vortex blocker size and its placement position. F. Measurement of Mixture Void Fraction. Instruments that can provide reliable void fraction measurement of the two-phase flow mixture are lacking. During this reporting period we continued the efforts to develop an instrument

	based on electric conductivity measurements. An analysis scheme has been developed to obtain the changes of magnitude and phase change with void fraction. In parallel, a light attenuation based measurement device is under development. Preliminary experiment indicates that both devices are good for void fraction estimations up to 20 ~ 30% gas volume fractions.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Wu X, Loraine G, Ma J, Hsiao C-T, Chahine GL. "Performance Studies of a DynaSwirl® Phase Separator for Space Applications." 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. , Nov-2013
Abstracts for Journals and Proceedings	Ma J, Hsiao C-T, Chahine GL. "Numerical Modeling of Bubbly Two-Phase Flow in a Vortex Separator." 9th International Conference on Two-Phase Systems for Ground and Space Applications, Baltimore, MD, September 22-26, 2014. 9th International Conference on Two-Phase Systems for Ground and Space Applications, Baltimore, MD, September 22-26, 2014. In press, as of July 2014. , Jul-2014